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Table of Contents

1. Introduction and Objective	2
2. Architecture & Justification	3
2.1 Choice of Topology	3
2.2 Detailed Justification	3
3. Manual Calculations	3
3.2 Defining Constraints	3
3.2 Stage 1	4
3.3 Stage 2	5
3.4 Stage 1	6
3.5 Total Gain Calculation	6
4. Simulations and Corrections	6
4.1 Original Idea (Pre-Correction/Changes)	6
4.2 New Idea (Post-Correction/Changes)	8
4.3 Design Discussion, Errors and Tradeoffs	9
5. Conclusion	10

1. Introduction and Objective

The aim of this project design is to design a simple analog voltage amplifier, an open-loop amplifier, based on Global Foundries 180 nm CMOS process specification using NMOS and PMOS. This amplifier design is based on architectural considerations and device-level compromises to fulfill a certain set of electrical requirements. The circuit comprises of three stages of gain using common-source configuration to obtain a total open-loop gain of $\geq 60dB$, followed by a common drain (source follower) configuration to tackle load dependence issues and output resistance problems. Some of the critical challenges of the design are keeping a total DC power consumption of $\leq 1mW$ while ensuring an output swing $\geq 1.5V_{pp}$ across a total $10k\Omega$ load.

The main goal of this project is to design and verify a MOSFET amplifier that follows the following technical specifications:

- Voltage Gain: Open loop gains $\geq 60dB$, Per stage gains $\leq 40dB$
- Power Efficiency: DC power consumption $\leq 1mW$, Branch currents $\leq 200\mu A$
- Signal Integrity: Output Voltage Swing $\geq 1.5V$, maintain $-3dB$ bandwidth $\geq 500kHz$
- Load Management: Gain reduction $\leq 10\%$ with $10k\Omega$ resistor and $2pF$ capacitor
- Input impedance: Input resistance $\geq 100k\Omega$

2. Architecture & Justification

2.1 Choice of Topology

Stage	Config.	Type	Justification
1	Common-Source (CS)	NMOS	Provide initial amplification.
2	Common-Source (CS)	NMOS	Provide further amplification.
3	Common-Drain (CD)	NMOS	Act as a buffer between gain/load.

Table 2.1.1. Table of topology and reasons for justification.

2.2 Detailed Justification

Originally it was decided that CS configuration would be the best for stage 1 as the gate of a MOSFET draws no DC current, allowing us to set the input impedance precisely as required by the lab (above 100k). Additionally, for stage 2, another CS configuration was selected in order to reach the target of 60dB. A single CS stage would not be able to accomplish this as there is a 40dB gain per stage limit as per the specifications. Lastly, a CD stage was chosen to satisfy the load sensitivity requirement, where the gain reduction could not be greater than 10%. A CD stage provides near unit voltage gain but high current gain, which would help maintain signal integrity.

3. Manual Calculations

3.2 Defining Constraints

Define constraints:
- Total power limit $\leq 1\text{ mV}$ at $V_{DD} = 3.3\text{V}$
- Total current: $1\text{mV}/3.3\text{V} = 303\mu\text{A}$
- Max branch current $\leq 200\mu\text{A}/\text{branch}$
↳ To stay safe, allocate 80-90 μA for each stage.

Figure 3.2.1. Defining constraints

3.2 Stage 1

Stage 1 Design:

- Lets take this to be common source
 - ↳ Provides voltage gain and high input resistance required. ($>100k\Omega$)
- since gate draws no DC current,
 - ↳ R_{in} is determined by biasing resistors R_1, R_2 .
 - ↳ To meet requirement, $R_1 \parallel R_2 > 100k\Omega$
- Lets choose $V_{ov} = 0.2V$
 - ↳ Within range (0.2-0.6)
 - ↳ low V_{ov} = large $g_m \rightarrow$ Easier to hit 60dB gain w/o exceeding power.
- Assume NMOS
- Given NMOS parameters:
 - ↳ $V_{tn} = 0.63V$
 - ↳ $\mu_n C_{ox} = 100\mu A/V^2$
 - ↳ Target $I_{D1} = 90\mu A$
- Calculating W/L

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{ov}^2$$

$$90\mu A = \frac{1}{2} (100\mu A/V^2) \left(\frac{W}{L}\right) (0.2V)^2$$

$$\frac{W}{L} = \frac{90}{50 \cdot 0.04} = 45$$

Figure 3.2.2. Calculations for Stage 1 CS.

- Calculate g_m :

$$g_m = \frac{2I_D}{V_{ov}} = \frac{2(90\mu A)}{0.2V} = 900\mu A/V = 0.9mS.$$
- Determining W & L

Tech spec: $L \geq 0.28\mu m$

Design spec: $0.5\mu m \leq L \leq 5\mu m$

Lets choose $L = 1\mu m$. Choosing one too little increases $CLM(A)$
 Choosing max ($5\mu m$) introduces parasitic capacitance which kill 50kHz bandwidth.

$$\therefore 45 = \frac{W}{L} \quad \begin{cases} W = 45\mu m \\ L = 1\mu m \end{cases}$$
- Calculating Drain resistor
 - ↳ Lets choose a value that results in a high stage gain $\leq 40dB$
 - $A_v \approx -g_m \cdot R_D$
 - ↳ Lets say $\sim 20V/V$ gain (26dB)
 - $20 = (0.9mS) \cdot R_D$
 - $R_D \approx 22.2k\Omega$
- Biasing the gate (V_{in})
 - $V_{GS} = V_{tn} + V_{ov} = 0.63 + 0.2 = 0.83V$
 - ↳ Therefore R_1 & R_2 must create 0.83V from provided 3.3V.

Figure 3.2.3. Continued calculations for stage 1.

- Calculating R_1, R_2
 (We now have two equations)

$$V_G = V_{DD} \cdot \left(\frac{R_2}{R_1 + R_2} \right) = 0.83V$$

$$R_{in} = R_1 \parallel R_L = \frac{R_1 \cdot R_2}{R_1 + R_2} \gg 100k\Omega \rightarrow \text{for safety, assume } R_{in} = 150k\Omega$$
 By the voltage ratio $\frac{0.83}{2.25} = 0.25$, we know $R_2 \approx \frac{1}{4} R_{Total}$
 Assume $R_1 = 600k\Omega$
 $R_2 = 200k\Omega$
 Check:

$$2.2 \cdot \frac{200}{800} = 0.825V \text{ (Close enough to } 0.83V)$$

$$\frac{(600 \cdot 200)}{(600 + 200)} = 150k\Omega \text{ (satisfies } \gg 100k\Omega)$$

Figure 3.2.4. Continued calculations for stage 1.

3.3 Stage 2

Stage 2 Design

- Let's take this to be common source again but this time coupling.
 (The output of stage 1 is high DC; cannot be directly connected to gate of stage 2.)
- To reach a total gain of 1000 V/V (60dB), and assuming final output buffer has gain 0.9 V/V, stage 2:

$$A_{V2} = \frac{1000}{20 \cdot 0.9} = 55 \text{ V/V}$$
- Assume NMOS again
 (Use same I_D and V_{ov} for same reasons)
- Calculating W/L , W & L
 $\frac{W}{L} = 45$ (from previous step)
 $L = 2\mu m \rightarrow$ setting it to $2\mu m$ gives lesser CLM (λ)
 $W = 2 \cdot 45 = 90\mu m$
- Calculating gain
 $A_{V2} = -g_{m2}(R_{D2} \parallel r_{o2})$
 g_{m2} is also $900\mu A$
 $r_{o2} = \frac{1}{\lambda \cdot I_{D2}}; \lambda = 0.04 \text{ (from guidelines)} \approx \frac{1}{0.04 \cdot 90\mu A} = 277k\Omega$
 (If we target a gain of 50 V/V
 $50 = (900\mu A/V) \times R_{Total}$
 $R_{Total} = 55.5k\Omega$
 $R_{Total} = R_{D2} \parallel r_{o2}; R_{D2} = 70k\Omega$

Figure 3.2.5. Calculations for stage 2 CS.

3.4 Stage 1

Stage 3 Design

- We will use a common-drain stage which will act as a buffer.
- This is a solution to the load sensitivity requirement (gain reduction $\leq 10\%$) and the output swing requirement ($\geq 1.5 V_{pp}$)
- Low output impedance ($R_{out} \approx 1/g_m$) to drive $10k\Omega$ load with minimal loss
- Use remainder of current budget.
 - say $120\mu A$ for this stage
 - $90 + 90 + 120 = 300\mu A$
 - $300\mu A \cdot 3.3V = 0.99mW \leq 1mW$ (satisfied!)
- Use $V_{ov} = 0.2V$ again to maximize g_m
- Dimensions W/L :

$$\frac{W}{L} = \frac{2 \cdot I_D}{\mu C_{ox} \cdot V_{ov}^2} = \frac{2 \cdot 120}{100 \cdot (0.2)^2} = 60$$

$$L = 0.5\mu m \text{ for bandwidth, } W = 30\mu m$$
- Transconductance:

$$g_m = \frac{2I_D}{V_{ov}} = \frac{240\mu A}{0.2V} = 1.2mS$$
- R_{out} :

$$R_{out} \approx \frac{1}{g_{m3}} = \frac{1}{1.2mS} = 833\Omega$$
- Gain with load:

$$A_{v3} = \frac{R_L}{R_L + (1/g_{m3})} = \frac{10,000}{10,000 + 833} = 0.923V/V$$

$\therefore$ There is a gain loss of $7.7\% \leq 10\%$ (satisfied!)

Figure 3.2.6. Calculations for Stage 3 CD.

3.5 Total Gain Calculation

$$A_V = A_{V1} \cdot A_{V2} \cdot A_{V3}$$

$$A_V = 60 \cdot 20 \cdot 0.923$$

$$A_V = 1107.6V/V \text{ or } 60.89dB$$

4. Simulations and Corrections

4.1 Original Idea (Pre-Correction/Changes)

The original idea upon implementation on KiCAD using SPICE software, brought the following results.

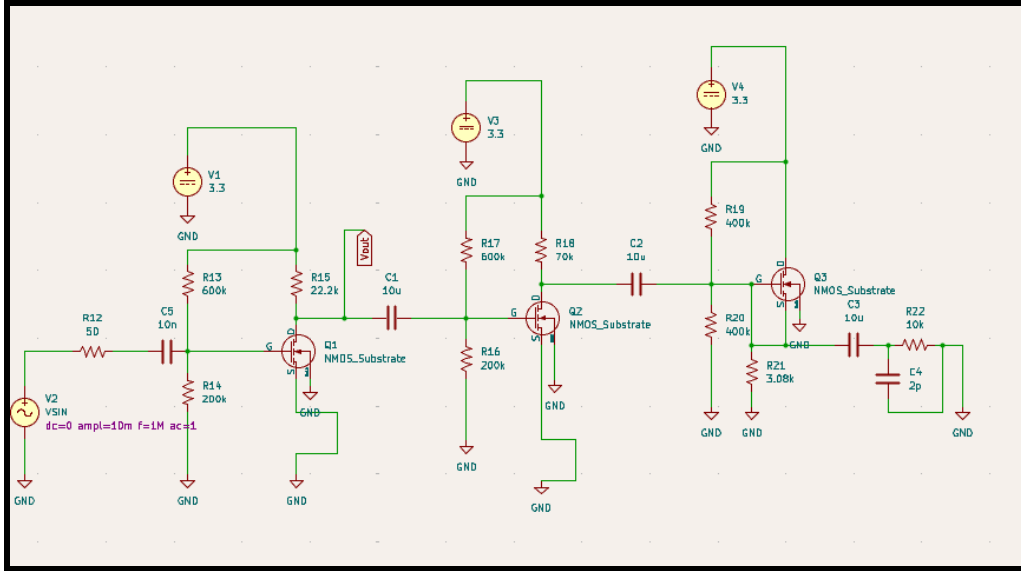


Figure 4.1.1. Schematic diagram of the designed topology.

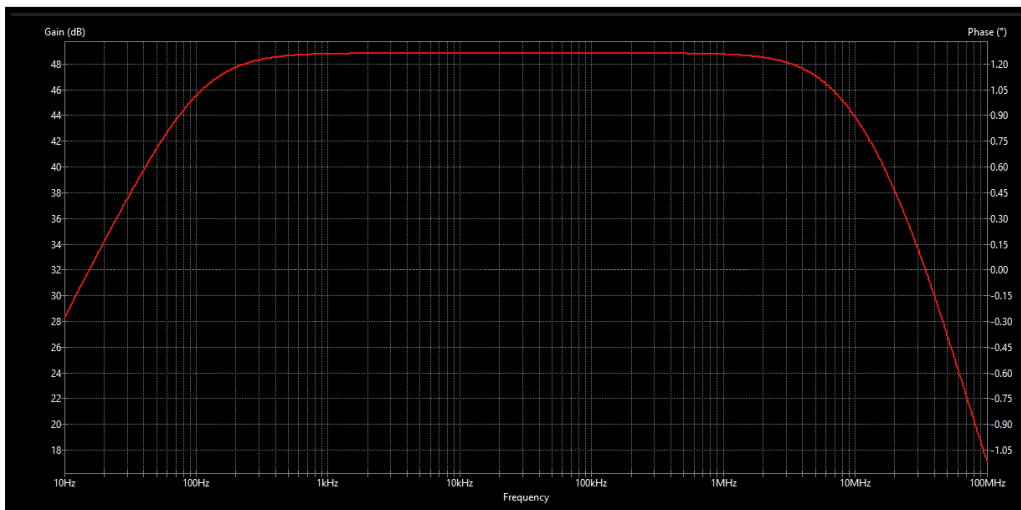


Figure 4.1.2. AC Analysis of the designed topology.

Upon completion of the testing process for the designed topology, it was clear that the 2 CS stages are incapable of producing enough amplification to get the desired gain of 60dB or higher. It was also understood that removing the CD stage and replacing it with a CS is not an option, since the gain reduction would for sure increase beyond the 10% threshold. Therefore, the next best step would be to add a brand new CS step before the buffer to help push the amplifier beyond the 60dB margin. Calculating the values for this new stage was not a hard deal, since we already knew a lot of information upon simulating the last design. Since we already had the current consumed by the 3 stages, we could allocate the rest of the current to this new CS stage. Based on the current, the rest of the values were found using the same process as the CS stage in stages 1 and 2. The values came out to be what is displayed in the figures.

4.2 New Idea (Post-Correction/Changes)

The original idea upon implementation on KiCAD using SPICE software, brought the following results.

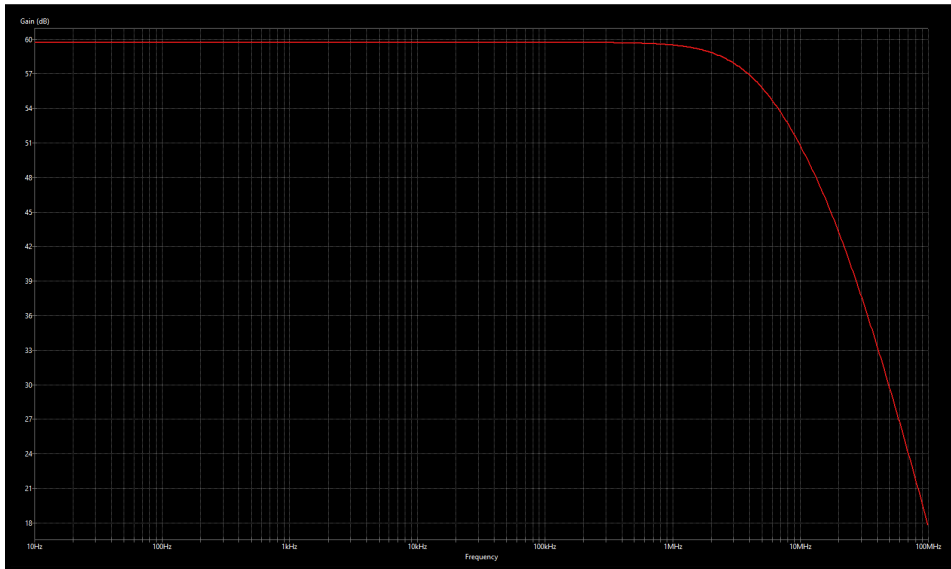


Figure 4.2.1. AC Analysis of the new topology.

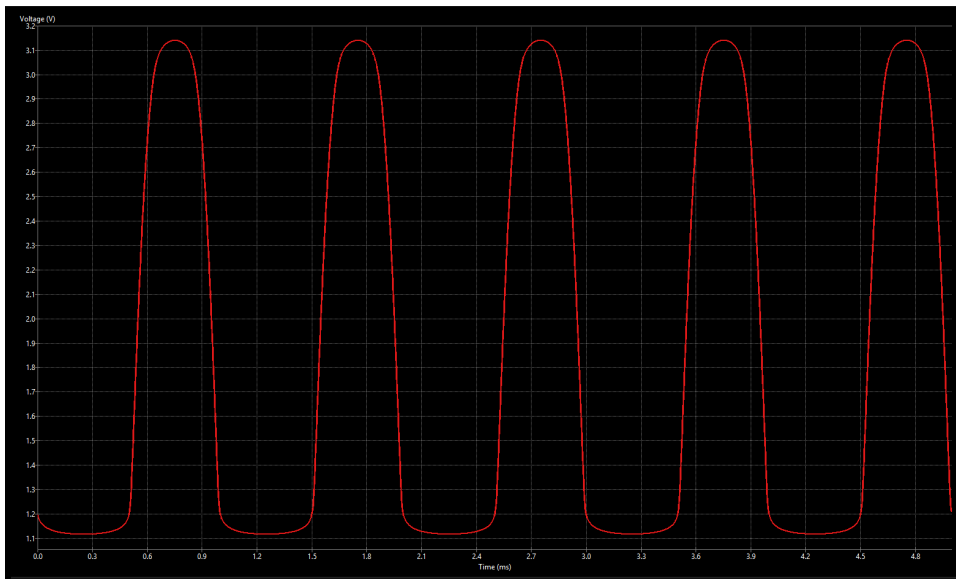


Figure 4.2.2. Transient Analysis of the new topology

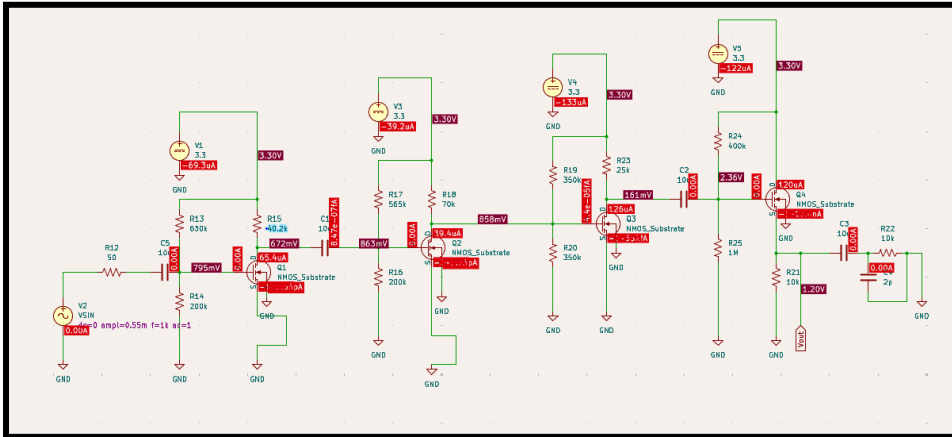


Figure 4.2.3. DC Operating Point and Schematic of new topology.

4.3 Design Discussion, Errors and Tradeoffs

The three graphs below represent the final empirical proof of success in the design, whereby the theoretical hand calculations have proved to translate to practical results of an effective high-gain amplifier circuit. The graph representing DC Operating Point verifies the successful operation of the biasing circuit in distributing the 1mW of power among the branches, whereby the total consumption amounts to about 0.87mW and all currents such as 126uA on the Q3 branch do not exceed 200uA. AC Sweep perfectly matches the 60dB required gain, whereas Transient Analysis proves the successful operation by having a good output signal swing of 2.0Vpp, more than enough when compared to the 1.5Vpp specified. The difference in values between hand calculation gain for each stage and simulation result gain can be attributed to effects such as Channel Length Modulation and body effect, both being ignored in hand calculations but precisely accounted for in SPICE simulation. Additionally, parasitic capacitance, affecting bandwidth and gain, causes the finite bandwidth seen here, contrary to ideal conditions.

In designing a high-quality amplifier subject to a very tight power budget of only 1 mW, there will be continuous tradeoffs involving gain, bandwidth, and swing. The key tradeoff that was encountered in this project design is between Gain and Power Efficiency, as since the transconductance (g_m) is current-dependent, it was necessary for us to have only around 40 to 60 μA in the gain stage circuit to adhere to the 1 mW power restriction, thus requiring a larger load resistance to attain the 60 dB requirement. This results in the Gain-Bandwidth tradeoff, where in increasing the load resistance to increase the gain, the dominant pole is also moved towards lower frequencies. However, this problem was overcome through the small size of the transistors used under the 180nm process, allowing for smaller parasitics, which kept the bandwidth to the desired value of 2 MHz. In addition, the transition from three to four gain stages constitutes an important tradeoff with regards to Linearity and Headroom. Through having the gains distributed across three Common Source stages, as opposed to two, the required voltage swing for each

individual transistor was lessened to avoid operation in the triode region. Lastly, a trade-off was done on the Current Allocation method where almost half of the current budget (120 μA) was spent on the Common-Drain amplifier. Although this sacrificed some currents from being amplified, it was necessary in decreasing the output resistance to meet the Load Sensitivity criterion of 10% at 10 k Ω load.

5. Conclusion

Conclusively, this design project was able to achieve a MOSFET amplifier design in four stages following the 180nm technology, and it satisfied all laboratory design criteria. The shift from a three-stage design to the CS-CS-CS-CD design was a key step in distributing the 60 dB gain in three different gain stages. This design strategy was important since it ensured that none of the stages would enter into the triode region when driving the CD buffer to meet the 1.5Vpp output swing. Hand calculations were done for the transconductance, g_m , and output resistance, r_o and this helped to provide a good starting point, which was further confirmed by the SPICE simulation. The findings showed that the target design can meet the 1mW power requirement without compromising performance and sensitivity to the load. In spite of challenges brought about by short channel effects and parasitic capacitances, the designed circuit had flat gain of 60 dB and bandwidth of 2 MHz.